

AI511/MM505 Linear Algebra with Applications

Lecture 9 – Coordinates, Dimension, Change of Basis

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Outline

Recap

Euclidean vector spaces

- Coordinates

- Dimension

- Change of basis

Recap

Linear combination and span

Definition 1.1.3 of Johnston (2021)

A *linear combination* of the vectors $v_1, \dots, v_k \in \mathbb{R}^n$ is any vector of the form

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k,$$

where $c_1, c_2, \dots, c_k \in \mathbb{R}$.

Definition 2.3.3 of Johnston (2021)

If $B = \{v_1, \dots, v_k\}$ is a set of vectors in $\mathbb{R}^n$, then the set of all linear combinations of those vectors is called their *span*, and it is denoted by $\text{span}(B)$ and $\text{span}(v_1, \dots, v_k)$.

Linear dependence and independence

Definition 2.3.4 of Johnston (2021)

A set of vectors $S = \{v_1, v_2, \dots, v_k\}$ is *linearly dependent* if there exist scalars $c_1, c_2, \dots, c_k \in \mathbb{R}$, at least one of which is not zero, such that

$$c_1 v_1 + \dots + c_k v_k = 0.$$

If S is not linearly dependent then it is called *linearly independent*.

- A set $B = \{v_1, \dots, v_k\}$ is called a basis of a subspace $\mathcal{S} \subset \mathbb{R}^n$ if it spans $\mathcal{S}$ and if it is linearly independent.
- If B is a basis and if we insert one more vector into B , $\mathcal{S}$ becomes linearly dependent.
- If B is a basis and if we remove one vector from B , B does not span $\mathcal{S}$ anymore.
- A basis is not too big and not too small—it is just right.

Euclidean vector spaces

Euclidean vector spaces

Coordinates

Coordinates

Definition 3.1.1 of Johnston (2021)

Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$, $B = \{v_1, v_2, \dots, v_k\}$ is a basis for $\mathcal{S}$, and $v \in \mathcal{S}$. Then the unique scalars $c_1, c_2, \dots, c_k$ for which

$$v = c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

are called the *coordinates* of v with respect to B , and the vector

$$[v]_B := (c_1, c_2, \dots, c_k)$$

is called the *coordinate vector* of v with respect to B .

When we work with coordinates or coordinate vectors, we simply use the basis vectors in the order written. If $B = \{v_1, v_2, \dots, v_k\}$, then the “first” basis vector is v_1 , the “second” basis vector is v_2 , and so on.

Examples of coordinates

- For $v = (v_1, \dots, v_n) \in \mathbb{R}^n$, we have

$$[v]_B = (v_1, \dots, v_n)$$

with respect to the standard basis $B = \{e_1, \dots, e_n\}$.

- The coordinates of $v = (5, 0)$ with respect to the basis

$$B' := \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

of $\mathbb{R}^2$ are $[v]_{B'} = (10, -5)$.

Euclidean vector spaces

Dimension

Number of vectors in a basis

Theorem 2.4.1 of Johnston (2021)

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ and suppose that $B, C \subset \mathcal{S}$ are finite sets with the properties that B is linearly independent and $\text{span}(C) = \mathcal{S}$. Then

$$|B| \leq |C|.$$

Linearly independent sets are “small”, while spanning sets are “big”.

Dimension

Corollary 2.4.2 of Johnston (2021)

Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$. Every basis of $\mathcal{S}$ has the same number of vectors.

Proof.

Suppose that B and C are bases of $\mathcal{S}$. Then $|B| \leq |C|$ since B is linearly independent and $\text{span}(C) = \mathcal{S}$. On the other hand, $|C| \leq |B|$ since C is linearly independent and $\text{span}(B) = \mathcal{S}$. Hence, $|B| = |C|$. $\square$

Definition 2.4.2 of Johnston (2021)

Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$. The number of vectors in a basis of $\mathcal{S}$ is called the *dimension* of $\mathcal{S}$ and is denoted by $\dim(\mathcal{S})$.

Examples of dimensions

- We formally define the dimension of $\{0\}$ to be 0.
- $\dim(\mathbb{R}^n) = n$.
- Lines are 1-dimensional since a single vector acts as a basis of a line.
- Planes in $\mathbb{R}^3$ are 2-dimensional since two non-parallel vectors form a basis of a plane.

Creating a basis

Theorem 2.4.3 of Johnston (2021)

Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$ and $B \subset \mathcal{S}$ is a finite set of vectors.

- (a) If B is linearly independent, then there is a basis C of $\mathcal{S}$ with $B \subset C$.
- (b) If B spans $\mathcal{S}$, then there is a basis C of $\mathcal{S}$ with $C \subset B$.

We can always toss away vectors from a spanning set until it is small enough to be a basis, and we can always add new vectors to a linearly independent set until it is big enough to be a basis.

Basis

If one already knows $\dim(\mathcal{S})$, then checking if something is a basis becomes easier.

Theorem 2.4.4 of Johnston (2021)

Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$ and $B \subset \mathcal{S}$ is a set containing k vectors.

- (a) If $k \neq \dim(\mathcal{S})$ then B is not a basis of $\mathcal{S}$.
- (b) If $k = \dim(\mathcal{S})$, then the following are equivalent
 - (i) B spans $\mathcal{S}$;
 - (ii) B is linearly independent;
 - (iii) B is a basis of $\mathcal{S}$.

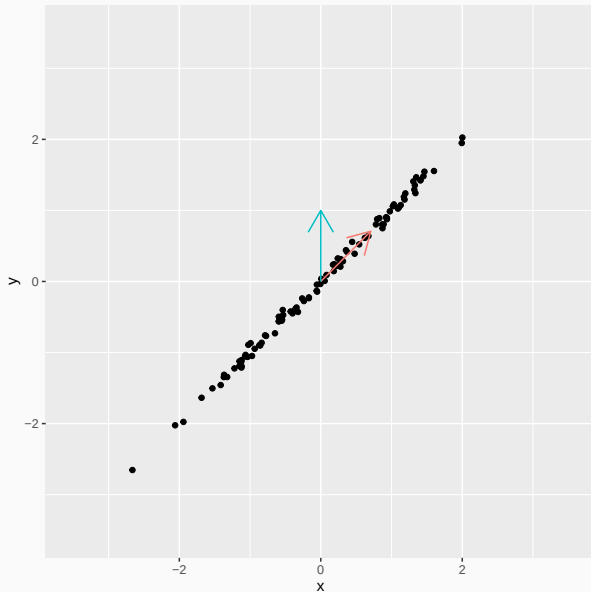
For instance, any two non-proportional vectors in $\mathbb{R}^2$ are a basis.

Proof.

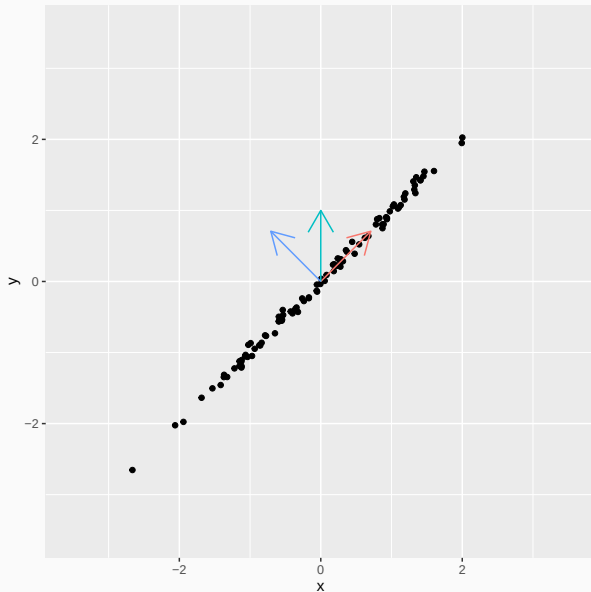
B spans $\mathcal{S} \implies B$ is linearly independent. Suppose otherwise, i.e., $\exists v \in B$ such that $v \in \text{span}(B \setminus \{v\})$. Then $\text{span}(B \setminus \{v\}) = \text{span}(B) = \mathcal{S}$. But this is a contradiction since no set with $k - 1$ elements can span a k -dimensional subspace.

B is linearly independent $\implies \text{span}(B) = \mathcal{S}$. Suppose otherwise, i.e., $\exists v \in \mathcal{S}$ such that $v \notin \text{span}(B)$. Then the set $B \cup \{v\}$ has $k + 1$ vectors and still is linearly independent. But this is a contradiction since a set with $k + 1$ vectors cannot be linearly independent in an k dimensional subspace (see Theorem 2.4.1 of Johnston (2021)). $\square$

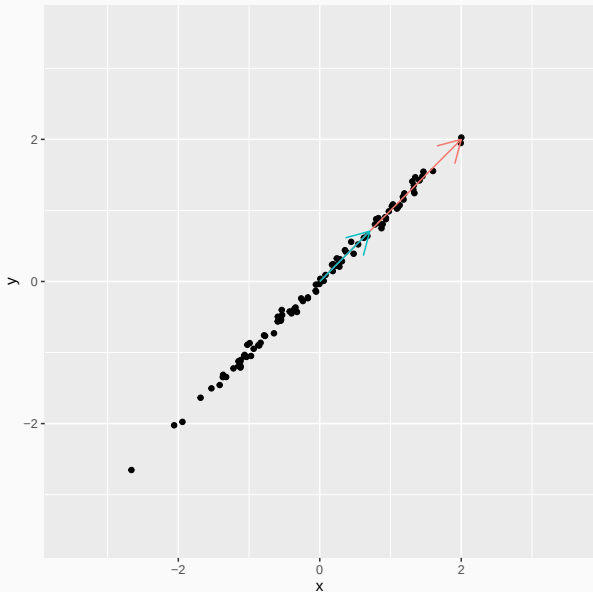
Are the two vectors a basis of $\mathbb{R}^2$?



Are the three vectors a basis of $\mathbb{R}^2$?



Are the two vectors a basis of $\mathbb{R}^2$ or of some subspace of $\mathbb{R}^2$?



Euclidean vector spaces

Change of basis

Example on the blackboard

Let

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \quad \text{and} \quad C = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

be bases of $\mathbb{R}^2$ and suppose that $[v]_B = (5, 1)$. Compute $[v]_C$.

Change of basis

- When we change the basis B that we are working with, the resulting coordinate vectors change as well.
- When we computed $[v]_C$ from $[v]_B$, we computed v itself as an intermediate step.
- This is quite undesirable in general since $[v]_B$ and $[v]_C$ can have a much lower dimension than v (consider a 3-dimensional subspace of $\mathbb{R}^{85}$).

Change-of-basis matrix

Definition 3.1.2 of Johnston (2021)

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ with bases $B = \{v_1, v_2, \dots, v_k\}$ and C . The *change-of-basis matrix* from B to C , denoted by $P_{C \leftarrow B}$, is the $k \times k$ matrix whose columns are the coordinate vectors $[v_1]_C, [v_2]_C, \dots, [v_k]_C$:

$$P_{C \leftarrow B} := \begin{bmatrix} [v_1]_C & [v_2]_C & \dots & [v_k]_C \end{bmatrix}.$$

Recommended to watch *Change of basis* of 3Blue1Brown.

Change of basis (cont.)

Theorem 3.1.3 of Johnston (2021)

Suppose B and C are bases of a subspace $\mathcal{S}$ of $\mathbb{R}^n$, and let $P_{C \leftarrow B}$ be the change-of-basis matrix from B to C . Then

- (a) $P_{C \leftarrow B}[v]_B = [v]_C$ for all $v \in \mathcal{S}$, and
- (b) $P_{C \leftarrow B}$ is invertible and $P_{C \leftarrow B}^{-1} = P_{B \leftarrow C}$.

Furthermore, $P_{C \leftarrow B}$ is the unique matrix with property (a).

- In part (a) of the theorem, the middle B s “cancel out” and leave just the C s behind (this is why the notation $P_{C \leftarrow B}$ is used instead of $P_{B \rightarrow C}$).
- One of the useful features of change-of-basis matrices is that they can be re-used to change the basis of multiple different vectors.

Change of basis (cont.)

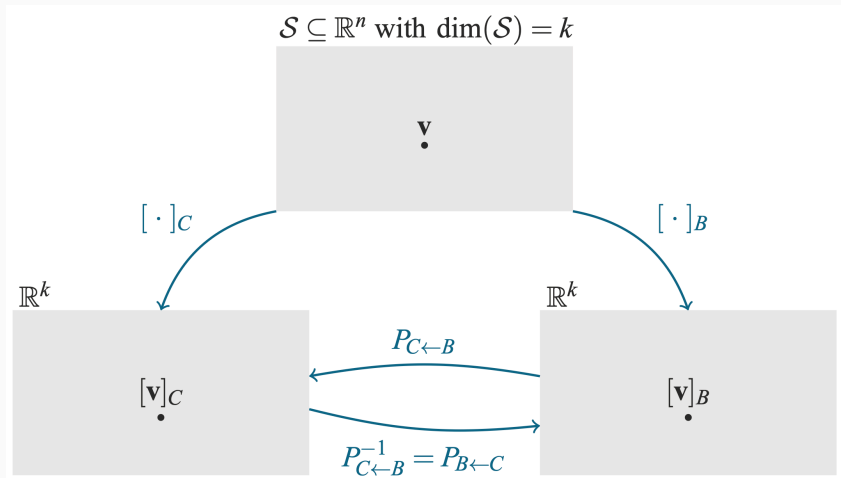


Figure 3.5 of Johnston (2021)

Example on the blackboard (cont.)

Let

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \quad \text{and} \quad C = \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

be bases of $\mathbb{R}^2$ and suppose that $[v]_B = (5, 1)$. Compute the change-of-basis matrix $P_{C \leftarrow B}$ and calculate $P_{C \leftarrow B}[v]_B$.

Recall that $[v]_C = (2, 2)$.



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.